```
vanam@Uday:~$ mkdir linux-day4
vanam@Uday:~$ cd linux-day4
vanam@Uday:~/linux-day4$
cat > file1.txt <<'EOF'
Java
Python
Linux
Spring Boot
SQL
EOF
vanam@Uday:~/linux-day4$
vanam@Uday:~/linux-day4$ cat > file2.txt <<'EOF'
Java
Python
Docker
Spring Boot
Git
EOF
vanam@Uday:~/linux-day4$ ls -ltr
total 8
-rw-r--r-- 1 vanam vanam 34 Oct 14 10:01 file1.txt
-rw-r--r-- 1 vanam vanam 35 Oct 14 10:01 file2.txt
vanam@Uday:~/linux-day4$
```

1. **1. diff — Compare two files**

Shows the differences between two files, including added, removed, or changed lines.

- `-u` — Unified format; shows changes with surrounding context.
- `-y` — Displays the files side by side.
- `-q` — Only reports whether the files differ.

```
vanam@Uday:~/linux-day4$ cat file1.txt
Java
Python
Linux
Spring Boot
SQL
vanam@Uday:~/linux-day4$ cat file2.txt
Java
Python
Docker
Spring Boot
Git
```

```
vanam@Uday:~/linux-day4$ diff file1.txt file2.txt
3c3
< Linux
---
> Docker
5c5
< SQL
---
> Git
```

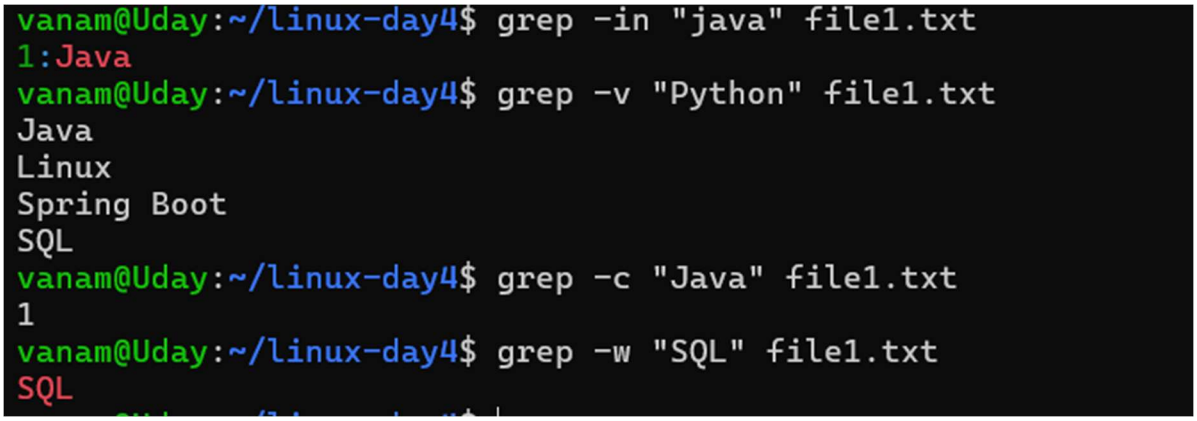
```
vanam@Uday:~/linux-day4$ diff -u file1.txt file2.txt
--- file1.txt   2026-10-14 10:01:14.335399747 +0000
+++ file2.txt   2026-10-14 10:01:22.947564641 +0000
@@ -1,5 +1,5 @@
 Java
 Python
-Linux
+Docker
 Spring Boot
-SQL
+Git
```

```
vanam@Uday:~/linux-day4$ diff -y file1.txt file2.txt
Java                                                            Java
Python                                                          Python
Linux                                                         | Docker
Spring Boot                                                     Spring Boot
SQL                                                           | Git
```

```
vanam@Uday:~/linux-day4$ diff -q file1.txt file2.txt
Files file1.txt and file2.txt differ
```

2. grep — Search for matching text

- `-i` — Ignore uppercase/lowercase differences.
- `-n` — Show line numbers.
- `-v` — Show lines that do not match.
- `-c` — Count matching lines.
- `-w` — Match a whole word.



3. sed — Edit or filter text

- `s/old/new/` — Replace the first occurrence on each line.
- `g` — Replace all occurrences on each line.
- `-n` — Suppress automatic printing.
- `p` — Print selected lines.



```
vanam@Uday:~/linux-day4$ sed -n '2,4p' file1.txt
Python
Linux
Spring Boot
```

```
vanam@Uday:~/linux-day4$ sed 's/Java/Advanced Java/g' file1.txt
Advanced Java
Python
Linux
Spring Boot
SQL
```

4. awk — Extract and process text

awk processes text by records and fields. It is useful for logs, CSV-like files, and command output.

- `$1` — First field.
- `$2` — Second field.
- `$NF` — Last field.
- `NF` — Number of fields in the current line.
- `NR` — Current input line number.
- `-F,` — Use a comma as the field separator.

```
vanam@Uday:~/linux-day4$ awk '{print $1}' file1.txt
Java
Python
Linux
Spring
SQL
```

```
vanam@Uday:~/linux-day4$ awk '{print NR, $1}' file1.txt
1 Java
2 Python
3 Linux
4 Spring
5 SQL
```

```
vanam@Uday:~/linux-day4$ awk '{print $NF}' file1.txt
Java
Python
Linux
Boot
SQL
```

5. sort — Sort lines of text

- `-r` — Reverse the sort order.
- `-n` — Sort numerically.
- `-u` — Remove duplicate lines from the sorted output.
- `-f` — Ignore case while sorting.
- `-o file` — Write output to a specified file.

```
vanam@Uday:~/linux-day4$ sort file1.txt
Java
Linux
Python
SQL
Spring Boot
```

```
vanam@Uday:~/linux-day4$ sort -r file1.txt
Spring Boot
SQL
Python
Linux
Java
```

```
vanam@Uday:~/linux-day4$ sort -f file1.txt
Java
Linux
Python
Spring Boot
SQL
```

6.comm — Find common and unique lines

Compares two sorted files and displays three columns:

1. Lines only in the first file.
2. Lines only in the second file.
3. Lines common to both files.

```
vanam@Uday:~/linux-day4$ comm file1.txt file2.txt
        Docker
        Git
                Java
Linux
                Python
SQL
                Spring Boot
```

- `-1` — Hide lines unique to the first file.
- `-2` — Hide lines unique to the second file.
- `-3` — Hide lines common to both files.

```
vanam@Uday:~/linux-day4$ comm -12 file1.txt file2.txt
Java
Python
Spring Boot
```

7. tr — Transform characters

```
vanam@Uday:~/linux-day4$ cat file1.txt | tr 'a-z' 'A-Z'
JAVA
LINUX
PYTHON
SQL
SPRING BOOT
```

- `-d` — Delete specified characters.
- `-s` — Squeeze repeated occurrences of a character into one.
- `-c` — Complement the specified character set.

```
vanam@Uday:~/linux-day4$ cat file1.txt | tr -d 'aeiou'
Jv
Lnx
Pythn
SQL
Sprng Bt
```

8. paste — Combine files side by side

Combines corresponding lines from multiple files, using a tab as the default separator.

```
vanam@Uday:~/linux-day4$ paste file1.txt file2.txt
Java    Docker
Linux   Git
Python  Java
SQL     Python
Spring Boot     Spring Boot
```

9. sha256sum — Verify file integrity

Calculates a SHA-256 checksum for a file. If the contents change, its checksum will normally change too.

```
vanam@Uday:~/linux-day4$ sha256sum file1.txt
sha256sum file2.txt
03aeee3284d661c28c2ea5e244e7bff1d314ae427084beeee1d34e99c9a04868  file1.txt
982bc6191354c720875c054a7cd21c6100473b232d8268dbc9630a08b3eb2542  file2.txt
```

To save and verify a checksum:

```
vanam@Uday:~/linux-day4$ sha256sum file1.txt > checksum.txt
sha256sum -c checksum.txt
file1.txt: OK
```

10. split — Divide a large file into smaller files

Splits a file into smaller pieces, which is useful for processing large logs or transferring files in chunks.

- `-l 2` — Put two lines in each piece.
- `-b 1K` — Split by size, approximately 1 KiB per piece.
- `-d` — Use numeric suffixes instead of alphabetic suffixes.

- `-a 3` — Set the suffix length to three characters.

